



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Forest Fire Risk Prediction Using Random Forest and Gradient-Boosted Trees

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF _____

Name of Title: Forest Fire Risk Prediction Using Random Forest and Gradient-Boosted Trees

1. Abstract:

Forest fires present a major threat to global ecosystems, human life, and property, requiring rapid early-warning mechanisms for proactive disaster management. Environmental factors such as temperature, relative humidity, wind speed, and fuel moisture indices play a critical role in determining fire risk and spread severity. This project implements a machine learning framework using ensemble models—specifically **Random Forest** and **Gradient-Boosted Trees**—to accurately predict forest fire risks.

Random Forest utilizes bootstrap aggregation to minimize variance and handle non-linear relationships, while Gradient Boosting sequentially optimizes loss functions to refine predictive performance. The methodology involves data preprocessing, feature selection, and model training using meteorological and Fire Weather Index (FWI) datasets. Models are evaluated across metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The findings demonstrate that ensemble tree methods effectively identify high-risk conditions, offering a reliable tool for early fire detection and forestry management.

Forest fires pose a critical threat to ecosystems and public safety, making early risk prediction essential for proactive disaster management. This project implements machine learning models—specifically **Random Forest** and **Gradient-Boosted Trees**—to predict forest fire risks using key meteorological metrics like temperature, humidity, wind speed, and fuel moisture indices. By evaluating dataset features and optimizing ensemble models across performance metrics such as Accuracy, Precision, and F1-Score, the framework provides a reliable early-warning mechanism to assist forestry teams in mitigating wildfire threats.



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2. INTRODUCTION:

Forest fires represent a growing global crisis, causing extensive destruction to biodiversity, economic infrastructure, and human lives. Traditional monitoring methods often rely on physical observation or simple thresholds, which struggle to capture the complex, non-linear relationships between dynamic weather conditions and fire ignition. Mitigating these disasters requires shifting from reactive firefighting strategies to data-driven, proactive prevention systems.

Machine learning provides a powerful approach for identifying patterns in environmental data. Among various algorithms, ensemble decision-tree models—specifically **Random Forest** and **Gradient-Boosted Trees**—are uniquely effective for handling spatial-temporal risk data. Random Forest reduces model variance by combining multiple independent decision trees through bagging. In contrast, Gradient-Boosted Trees sequentially minimize prediction errors to capture subtle variations in environmental conditions.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows 10/11 (64-bit), macOS, or Linux

Programming Language: Python 3.8+

Development Tools: Jupyter Notebook, Google Colab, or VS Code

Core Libraries & Frameworks:

- **Data Processing:** Pandas, NumPy
- **Machine Learning:** Scikit-Learn, XGBoost, LightGBM
- **Visualization:** Matplotlib, Seaborn

Signature of the Guide

Course Instructor